RESEARCH FINDINGS

Earth Systems Observatory — Thesis A

Severe Geomagnetic Storms and M7.0+ Subduction Zone Earthquakes

A 15-Year Statistical Investigation of Southward Interplanetary Magnetic Field and Large Seismic Events

Earth Systems Observatory | April 2026

Data: NASA OMNIWeb | USGS FDSNWS ComCat

Statistical method: One-sided binomial test | Aftershock-declustered catalog

Abstract

This study examines whether solar wind disturbances — specifically southward excursions of the Interplanetary Magnetic Field (IMF Bz) — are statistically associated with M7.0+ subduction zone earthquakes in the following 24–96 hours. Using 15 years of NASA OMNI hourly solar wind data (2011–2026) and a declustered USGS seismic catalog (4,798 mainshocks after removal of 2,124 aftershocks), we applied a one-sided binomial test against an empirical Poisson baseline rate.

Primary hypothesis (Bz ≤ −10 nT): Null result. Over 220 independent trigger events (10-day minimum spacing), the M7.0+ subduction zone earthquake rate shows no elevation ($p = 0.663$, RR = 0.93x). This supersedes a preliminary 5-year false positive ($p = 0.027$) that did not survive methodological improvement and extended data collection.

Exploratory finding (Bz ≤ −12 nT): Severe geomagnetic storms — 33 independent events over 15 years — are followed by M7.0+ subduction zone earthquakes at 3.6 times the baseline rate ($p = 0.0027$, surviving Bonferroni correction). The hit rate is 21.2% against a 5.9% Poisson baseline. The distribution of hits across the 24–96h window is bimodal, with significant clustering in the 24–48h window ($p = 0.028$, 3 events) and the 72–96h window ($p = 0.004$, 4 events), and a complete absence of hits in the intervening 48–72h period. Geographically, hits span 5 distinct subduction zones across the western Pacific and north Pacific arc, with no single zone accounting for more than two events.

This finding is **exploratory** — it emerged from a pre-planned sensitivity analysis rather than a pre-registered primary hypothesis. It generates a testable, falsifiable prediction for prospective evaluation: Bz ≤ −12 nT sustained ≥ 3h → M7.0+ subduction zone earthquake within 24–96h at greater than 5.9% baseline rate. The bimodal lag pattern, if confirmed, may implicate two distinct physical pathways operating at different phases of the geomagnetic storm cycle.

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1. Introduction

The relationship between solar activity and terrestrial seismicity has been a subject of periodic scientific interest since the early 20th century. Most proposed mechanisms involve the coupling of solar wind energy into the geomagnetic field — specifically, the role of southward-directed Interplanetary Magnetic Field (IMF Bz) in opening the magnetopause and driving geomagnetic storm activity. During storm main phases, large geomagnetically induced currents (GICs) flow through conductive Earth layers, and it has been hypothesized that these currents could modulate fault stress through electrokinetic or piezomagnetic coupling.

Prior statistical studies have produced contradictory results, largely due to small sample sizes, inconsistent catalog treatments (particularly the failure to remove aftershock sequences), and threshold-shopping across solar wind parameters. This investigation attempts to address these limitations with a systematic, methodologically transparent analysis that explicitly accounts for multiple comparisons, aftershock contamination, and trigger event independence.

The analysis proceeded in two stages. First, a 5-year pilot study (2021–2026) appeared to show a statistically significant elevation at M7.0+ for Bz ≤ −10 nT ($p = 0.027$, RR = 1.56x). This result was then subjected to a rigorous expert methodology audit that identified four major concerns: aftershock contamination, trigger non-independence, multiple comparison inflation, and solar

cycle confounding. A methodologically improved 15-year replication was conducted, and the pilot result did not survive. The extended analysis additionally included a pre-planned Bz threshold sensitivity sweep, which identified a stronger and statistically more robust association at Bz ≤ −12 nT.

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2. Data and Methods

2.1 Solar Wind Data

Hourly solar wind observations were obtained from the NASA OMNIWeb database (*omniweb.gsfc.nasa.gov*), which provides a merged, cleaned record from upstream L1 monitors including ACE and DSCOVR. Two parameters were extracted: bulk solar wind speed (km/s, variable 24) and IMF Bz in GSM coordinates (nT, variable 15). Data fill values (speed > 9990, |Bz| > 999) were excluded. The study window is 2011 March 13 to 2026 March 12 (131,496 hourly records), encompassing Solar Cycle 24 from near-maximum through minimum, and the ascending phase of Solar Cycle 25. This range was chosen specifically to span multiple solar cycle phases and reduce solar-cycle confounding.

2.2 Seismic Catalog

M5.5+ earthquake records were downloaded from the USGS FDSNWS ComCat service in 2-year chunks to remain within the 20,000-event API limit. The raw catalog contained 6,922 events (after deduplication at chunk boundaries).

Aftershock declustering was applied using a simplified Gardner-Knopoff (1974) algorithm. For each earthquake (potential mainshock), a space-time exclusion window was defined based on its magnitude: for M7.0+, the window extends 200 km and 365 days; for M6.5–6.9, 150 km and 200 days; for M6.0–6.4, 100 km and 90 days; for M5.5–5.9, 70 km and 60 days. Subsequent smaller events within these windows were flagged as aftershocks and removed. This procedure removed 2,124 events (30.7% of the raw catalog), leaving 4,798 independent mainshocks for analysis.

Subduction zone classification used a lookup table of 20 named zones defined by latitude, longitude bounding boxes. Events not falling within any zone boundary were excluded from outcome measurement but retained for baseline rate calculation.

2.3 Trigger Detection

A solar wind trigger was defined as a sustained exceedance of threshold conditions for ≥ 3 consecutive hourly records. Two conditions were evaluated independently: (1) bulk speed ≥ 600 km/s ("speed trigger"), and (2) IMF Bz ≤ threshold ("Bz trigger"). Triggers satisfying both conditions were classified "both"; those satisfying only one were classified accordingly.

To enforce event independence and avoid counting multiple triggers from the same geomagnetic storm sequence, a minimum inter-trigger gap of 240 hours (10 days) was applied: after any trigger, the algorithm advances at least 10 days before considering the next candidate. This reduced the effective trigger count from over 600 raw exceedances to 220 independent events — 142 speed-only, 69 Bz-only, and 9 both.

2.4 Outcome Measurement and Statistical Framework

For each trigger event, the outcome window was defined as 24–96 hours post-trigger. A trigger was counted as a "hit" if any M≥threshold subduction zone mainshock occurred within this window. Triggers were right-censored: any trigger whose 96-hour outcome window extended beyond the last available OMNI record was excluded from analysis.

The statistical baseline rate was computed using a Poisson model: the empirical rate of M≥threshold subduction mainshocks per hour was calculated from the full declustered catalog, converted to expected counts per 72-hour window (= LAG_MAX − LAG_MIN), and transformed to a probability via $P = 1 - e^{-\lambda t}$. A one-sided binomial test $P(X \geq k \mid n, p_{baseline})$ was applied, using exact enumeration for $n < 50$ and a continuity-corrected normal approximation otherwise.

For the magnitude sweep across four thresholds (M5.5, 6.0, 6.5, 7.0), a Bonferroni-corrected significance threshold of $p < 0.0125$ was applied. The Bz sensitivity analysis tested three Bz thresholds (−8, −10, −12 nT) and was pre-planned prior to the 15-year run.

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3. Results

3.1 Primary Hypothesis: Bz ≤ −10 nT — Null Result

Across 220 independent trigger events spanning 15 years, no statistically significant elevation in M7.0+ subduction zone earthquake rate was observed at any magnitude threshold. Results are shown in Table 1.

Table 1. Magnitude Sweep Results (15-year declustered catalog, all trigger types, n = 220)

Threshold	Quakes	Triggers	Hits	Hit Rate	Baseline	p-value	Rate Ratio
M5.5+	4,798	220	144	65.5%	72.3%	0.991	0.91x
M6.0+	1,559	220	76	34.5%	36.3%	0.731	0.95x
M6.5+	522	220	26	11.8%	14.4%	0.879	0.82x
M7.0+	194	220	12	5.5%	5.9%	0.663	0.93x

Rate ratios < 1.0 indicate no elevation. None of the four thresholds approach significance. Bonferroni threshold: p < 0.0125.

The trigger type breakdown at M7.0+ reveals that neither speed-only triggers ($p = 0.748$, RR = 0.84x) nor Bz-only triggers at −10 nT ($p = 0.413$, RR = 1.23x) reach significance. The preliminary 5-year signal has entirely dissolved with the expanded, methodologically corrected dataset.

3.2 Sensitivity Analysis: Severe Storm Signal at Bz ≤ −12 nT

The pre-planned Bz threshold sensitivity analysis tested whether the null result was consistent across nearby thresholds, or whether a subpopulation of more extreme events showed a different pattern. Results are shown in Table 2.

Table 2. Bz Threshold Sensitivity Analysis — Bz-Only Triggers at M7.0+

Bz Threshold	Bz-Only Triggers	Hits	Hit Rate	Baseline	p-value	Rate Ratio	
≤ −8 nT	126	8	6.3%	5.9%	0.490	1.08x	—
≤ −10 nT	69	5	7.2%	5.9%	0.413	1.23x	—
≤ −12 nT	**33**	**7**	**21.2%**	5.9%	**0.003**	**3.60x**	*****

** Survives Bonferroni correction (threshold p < 0.0125 for 4 magnitude tests). Exact p = 0.0027.*

Key Finding: Severe southward IMF events (Bz ≤ −12 nT sustained ≥ 3h) are followed by M7.0+ subduction zone earthquakes at 3.6× the baseline rate. Of 33 independent severe storm events over 15 years, 7 were followed by a qualifying earthquake within 96 hours — against an expectation of approximately 2 events by chance alone (p = 0.0027, one-sided binomial). This finding survives Bonferroni correction.

3.3 The Seven Hit Events

Table 3 lists all seven M7.0+ subduction zone earthquakes that occurred within 24–96 hours of a severe Bz ≤ −12 nT storm event, drawn from 15 years of independent trigger events (n = 33).

Table 3. Hit Events — M7.0+ Earthquakes Following Severe Geomagnetic Storms

#	Storm Date	Lag	Quake Date	Mag	Zone	Window
1	2015-03-28	**29h**	2015-03-29	**M7.5**	Solomon Islands Trench	24–48h
2	2015-05-06	**24h**	2015-05-07	**M7.1**	Solomon Islands Trench	24–48h
3	2016-01-21	**78h**	2016-01-24	**M7.1**	Alaska-Aleutian Trench	72–96h
4	2022-03-13	**73h**	2022-03-16	**M7.3**	Japan Trench	72–96h
5	2023-01-05	**83h**	2023-01-08	**M7.0**	New Hebrides Trench	72–96h
6	2023-07-14	**33h**	2023-07-16	**M7.2**	Alaska-Aleutian Trench	24–48h
7	2024-08-04	**83h**	2024-08-08	**M7.1**	Ryukyu Trench	72–96h

Lag = hours from storm onset to earthquake. Teal = 24–48h window; amber = 72–96h window.

3.4 Bimodal Lag Distribution

A striking feature of the severe-storm results is the distribution of hits across the 24–96h lag window. Table 4 shows that hits do not distribute uniformly — they concentrate strongly in two sub-windows with a complete gap between them.

Table 4. Lag Window Split — Bz ≤ −12 nT Severe Storm Triggers at M7.0+

Window	N Triggers	Hits	Hit Rate	Baseline	p-value	
24–48h	33	3	9.1%	2.0%	**0.028**	←
48–72h	33	0	0.0%	2.0%	1.000	
72–96h	33	4	12.1%	2.0%	**0.004**	←
24–96h (full)	33	7	**21.2%**	5.9%	**0.003**	←

Zero hits in the 48–72h window creates a bimodal distribution. Both the early (24–48h) and late (72–96h) sub-windows individually reach or approach significance.

The 48–72h null window — zero hits in 33 trigger events — is not a statistical artefact of small sample size. Under the Poisson model, the expected number of hits in this window is 0.66, giving a probability of observing zero hits of $e^{-0.66} \approx 0.52$. The observation is therefore unremarkable in isolation. However, when combined with the significant clustering on either side, it suggests a structured rather than uniform response. If the distribution were uniform across the full 72-hour outcome period, one would expect roughly equal counts in each 24-hour sub-window, not the observed 3–0–4 pattern.

3.5 Geographic Distribution

The seven hit events span five distinct subduction zones: Solomon Islands (2×), Alaska-Aleutian (2×), Japan Trench (1×), New Hebrides (1×), and Ryukyu (1×). All are in the western or north Pacific. While western Pacific dominance reflects the globally higher baseline seismicity of this region (approximately 50–60% of global subduction zone M7+ events), the geographic spread across five separate tectonic systems — rather than concentration at a single, persistently active zone — is consistent with a global forcing mechanism rather than local seismic cycle coincidence.

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4. Discussion

4.1 The False Positive and What It Teaches

The 5-year pilot result ($p = 0.027$ at $B_z \leq -10$ nT) is an important case study in small-N statistical fragility. Seven hit events over 60 triggers — with no aftershock declustering and inadequate trigger spacing — produced a nominally significant result that completely dissolved when analyzed properly. The lesson is not that solar-seismic coupling is implausible, but that the specific threshold of −10 nT is too broad: it includes moderate storms that appear to have no seismic association, diluting any signal from the most severe events. The audit process — examining the methodology before writing the document — was essential to catching this before the analysis was presented as established.

4.2 Interpreting the Bimodal Lag

The most structurally interesting result in this dataset is the 3–0–4 distribution of hits across the 24–48h, 48–72h, and 72–96h sub-windows. Several interpretations are worth considering.

Storm phase timing. A major CME-driven geomagnetic storm typically follows a characteristic time profile: sudden commencement (0–3h), initial phase (3–12h), main phase (12–48h during which Dst reaches minimum), and recovery phase (48–120h). The 24–48h hit cluster corresponds to the late main phase or Dst minimum — the period of maximum ring current intensity and therefore maximum GIC amplitude. The 72–96h cluster corresponds to the storm recovery phase, during which the ring current decays and GIC character changes. The 48–72h gap corresponds to the transition between these phases. This interpretation, while speculative, is internally consistent with known storm structure.

Two fault populations. An alternative interpretation is that two distinct populations of faults respond on different timescales: critically-loaded shallow megathrust patches that respond quickly to any perturbation (24–48h), and deeper or more locked segments that respond on the longer 72–96h timescale. Distinguishing these explanations would require focal mechanism and depth analysis of the seven events — data available in the USGS catalog for future work.

Sampling noise. With seven events total, the bimodal pattern could be a statistical accident. Under a null model of uniform distribution across the 72-hour window, the probability of observing a 3–0–4 or more extreme split is not vanishingly small. This possibility cannot be ruled out and is a primary motivation for prospective replication.

4.3 Physical Mechanism Considerations

The most significant challenge to this analysis is the proposed physical mechanism. Geomagnetically induced current amplitudes produce electromagnetic stress perturbations on fault systems estimated at 10^{-3}–10^{-1} Pa — orders of magnitude smaller than tidal loading (10^{2}–10^{3} Pa), which itself shows only marginal seismic triggering effects. Any proposed mechanism must account for this energy scaling problem.

One avenue that has not been fully explored is whether the relevant variable is not the steady-state GIC amplitude but rather the rate of change — transient current spikes during storm sudden commencement or substorm injections that may briefly exceed the time-averaged values by one to two orders of magnitude. Another is whether the effect is nonlinear, with a meaningful threshold below which no triggering occurs and above which the probability of triggering increases sharply. The observed threshold-dependence (−10 nT: no effect; −12 nT: 3.6× effect) is qualitatively consistent with a nonlinear response.

The finding that speed-only triggers (CME/CIR ram pressure without significant southward Bz) show no elevation is an important specificity result. Ram pressure changes and southward Bz co-occur frequently but are physically distinct: the former affects the magnetosphere through momentum transfer, the latter through magnetic reconnection. If the signal were due to generic solar wind perturbation, speed triggers should show similar elevation. Their absence implies the electromagnetic channel — ring current development, GIC — rather than pressure loading is the relevant pathway, if any pathway exists.

4.4 Limitations

Several limitations should be noted. First, while aftershock declustering was applied, the Gardner-Knopoff parameters used are calibrated primarily on southern California seismicity and may not perfectly match global subduction zone scaling. Second, the USGS catalog appears to return approximately 450–500 M5.5+ events per year against an expected global rate of ~1,500–2,000, suggesting geographic or magnitude-completeness filtering may be affecting the absolute counts. Since the baseline rate is computed from the same catalog, relative comparisons are internally valid, but this warrants further investigation. Third, the analysis does not control for ENSO phase, which modulates western Pacific seismicity and could act as a confounder if ENSO state correlates with solar activity through an indirect pathway. Fourth, the 10-day trigger spacing, while conservative, may still allow some correlated storm pairs within active CME periods.

4.5 Path to Pre-Registered Replication

The exploratory status of the −12 nT finding means it cannot be treated as established evidence. The appropriate next step is a pre-registered replication with the following specifications, agreed in advance of data collection:

1. Primary hypothesis: Bz ≤ −12 nT sustained ≥ 3h ("severe Bz trigger") is followed by M7.0+ subduction zone mainshock within 24–96h at greater than 5.9% baseline rate.
2. Primary test: one-sided binomial $p < 0.05$ (single pre-specified test, no multiple comparison adjustment needed).

3. Minimum sample: 25 independent severe Bz trigger events post-registration (estimated 2–3 years of prospective monitoring given observed frequency of ~2.2 events/year).
4. Pre-specified secondary test: whether hits are distributed non-uniformly across the 24–48h and 72–96h sub-windows (chi-squared test against uniform distribution).
5. Aftershock declustering must be applied to the outcome catalog using the same parameters as this study.
6. Trigger event independence: 10-day minimum spacing enforced.

5. Conclusion

This 15-year statistical investigation found no support for the hypothesis that moderate geomagnetic disturbances (IMF Bz ≤ −10 nT) are associated with elevated M7.0+ subduction zone seismicity. A preliminary 5-year result ($p = 0.027$) did not survive methodological improvement, extended data collection, or aftershock declustering — illustrating the importance of rigorous pre-analysis methodology review.

A pre-planned sensitivity analysis identified a potentially significant association at a higher severity threshold: severe southward IMF events (Bz ≤ −12 nT sustained ≥ 3h) are followed by M7.0+ subduction zone earthquakes at 3.6 times the background rate ($p = 0.0027$, $n = 33$ independent events, Bonferroni-corrected). Seven specific events support this finding, spanning five subduction zones across a 9-year period. The lag distribution is bimodal, with hits clustering in the first 48 hours and again in the 72–96h window, with a complete absence of hits in the intervening 48–72h period.

> **This is an exploratory finding, not established evidence.** It generates a specific, falsifiable, pre-registerable hypothesis: severe geomagnetic storms (Bz ≤ −12 nT) trigger M7.0+ subduction zone earthquakes within 96 hours at elevated rates. The bimodal lag structure, the specificity to Bz rather than speed triggers, and the geographic spread of hit events are consistent with but do not establish an electromagnetic forcing mechanism. Prospective pre-registered replication over 2–3 years of monitoring is the recommended next step.

Thesis A — Severe Geomagnetic Storms and Subduction Zone Seismicity

Addendum A1: Focal Mechanism and Depth Analysis of Seven Hit Events

Motivated by Perplexity AI peer review, April 2026 | Earth Systems Observatory Project

1. Introduction and Motivation

The primary Thesis A analysis (Bz ≤ −10 nT, n = 220 triggers) returned a null result (RR = 0.93×, p = 0.663). The exploratory Bz ≤ −12 nT analysis identified seven hit events across five subduction zones (p = 0.0027, Bonferroni-corrected; RR = 3.6×), with a bimodal lag distribution: three hits clustered in the 24–48 h window and four in the 72–96 h window, with zero hits in the intervening 48–72 h gap.

A subsequent independent review (Perplexity AI, April 2026) flagged this bimodal structure as the most analytically interesting feature in the dataset and recommended examining focal mechanisms and depths of the seven hit events as a means of probing the physical origin of the two distinct lag clusters. This addendum reports that analysis.

The core question is whether the early-window events (24–48 h, n = 3) and late-window events (72–96 h, n = 4) are distinguishable by any systematic seismic property—depth, fault mechanism, storm intensity, or tectonic zone—that would either support or argue against a causal geomagnetic trigger.

2. Data and Methods

2.1 Event Identification

The seven hit events were identified in the Thesis A v2.0 post-audit backfill (backfill-thesis-a.py, 15-year span, Gardner–Knopoff declustered, 10-day minimum trigger gap). Event date, magnitude, zone assignment, storm peak Bz, and lag time in hours were drawn directly from the severeBzDeepDive section of thesis-a-backfill.json.

2.2 Focal Mechanism and Depth Sources

Focal mechanism and depth data were retrieved from multiple authoritative sources: the USGS ANSS ComCat (event pages for USGS IDs us6000h519 and us7000kg30), the Alaska Earthquake Center (Sand Point M7.2), the Earthquake Insights network analysis (Ryukyu M7.1), Watchers News seismological reporting (Vanuatu M7.0), Air Worldwide Post-Event Analysis (Fukushima M7.3), and published academic literature for the 2015 PNG/Solomon Islands doublet. Where depth estimates varied between agencies, the USGS preferred depth is used; discrepancies are noted.

Fault type classification follows standard GCMT rake-angle conventions: thrust (rake 45–135°), normal (rake −135 to −45°), and strike-slip (rake within ±45° of 0° or 180°). Where moment tensor rake angles are not reported in available sources, fault type is assigned from regional tectonic context and published descriptions of the event.

2.3 Analytical Approach

Given n = 7, no formal statistical tests for group differences are attempted; the sample is too small for reliable inference. The analysis is descriptive and hypothesis-generating only. Three properties are examined for systematic variation between lag windows: (1) hypocentral depth, (2) fault mechanism type, and (3) peak storm Bz at trigger time. A fourth examination considers tectonic zone diversity across windows.

3. Seven Hit Event Catalogue with Focal Parameters

Table 1 presents all seven hit events with storm parameters, seismic parameters, and focal mechanism data. Events are ordered by lag time. The lag window assignment (Early = 24–48 h; Late = 72–96 h) is indicated in the final column.

#	Storm Date	Peak Bz (nT)	M	Zone	Depth (km)	Mechanism	Lag & Window
2	2015-05-06	−15.6	7.1	Solomon Islands Trench	~18	Thrust — shallow subduction interface (PNG/Solomon arc)	**24.2 h • EARLY**
1	2015-03-28	−13.9	7.5	Solomon Islands Trench	~10–15	Thrust — shallow subduction interface (PNG/Solomon arc)	**28.8 h • EARLY**
6	2023-07-14	−13.2	7.2	Alaska-Aleutian Trench	9–25	Thrust — subduction zone interface (Sand Point megathrust)	**32.8 h • EARLY**
4	2022-03-13	−15.5	7.3	Japan Trench	41–57	Thrust — intra-slab reverse fault within subducting Pacific plate	**72.6 h • LATE**
3	2016-01-21	−15.4	7.1	Alaska-Aleutian Trench	60–80†	Strike-slip — intra-slab, within subducted Pacific plate†	**77.5 h • LATE**
5	2023-01-05	−15.5	7.0	New Hebrides Trench	10–28	Thrust — megathrust interface or intra-plate thrust, Vanuatu arc	**82.5 h • LATE**
7	2024-08-04	−16.7	7.1	Ryukyu Trench	25	Thrust — Philippine	**82.7 h • LATE**

					Sea plate subduction megathrust interface

Table 1. *Seven hit events from the Bz ≤ −12 nT exploratory analysis, sorted by lag time. Blue rows = early-window (24–48 h); yellow rows = late-window (72–96 h). † The 2016 Alaska event (Hit 3) is located near Anchor Point / Cook Inlet rather than the outer Aleutian arc; its strike-slip intra-slab mechanism distinguishes it from the other six thrust events. Depth ranges reflect variation in estimates across agencies.*

4. Systematic Analysis of Lag-Window Associations

4.1 Hypocentral Depth

Depth is the most physically motivated candidate for explaining the bimodal lag. If the triggering mechanism involves stress propagation from the surface (e.g., via GIC-induced lithospheric currents or secondary strain waves), deeper events might plausibly respond on longer timescales.

Early-window events (24–48 h):

> Hit 2 (Solomon Is.): ~18 km | Hit 1 (Solomon Is.): ~10–15 km | Hit 6 (Sand Point, AK): 9–25 km

Mean depth: ~15–19 km — uniformly shallow subduction-zone events.

Late-window events (72–96 h):

> Hit 4 (Japan): 41–57 km | Hit 3 (Alaska): ~60–80 km | Hit 5 (Vanuatu): ~10–28 km | Hit 7 (Ryukyu): 25 km

Mean depth: ~35–45 km — notably deeper on average, but with Vanuatu and Ryukyu as shallow outliers.

Directional trend: there is a suggestive depth pattern. The two deepest events in the dataset (Japan 41–57 km; Alaska 60–80 km) both fall in the late window, while the three shallowest events (all <25 km) occupy the early window. Vanuatu (~10–28 km) is the principal exception—a shallow event in the late window. With n = 7 this cannot be considered statistically meaningful, but it is consistent with a depth-mediated delay hypothesis and should be examined again once 25+ prospective triggers are available.

4.2 Fault Mechanism Type

Six of the seven hit events have thrust mechanisms, consistent with their subduction zone setting. The single exception is Hit 3 (Alaska, January 2016), which is an intra-slab strike-slip event at intermediate depth (~60–80 km) within the subducted Pacific plate in the Cook Inlet / Kenai Peninsula area, distinct from the outer Aleutian arc megathrust. This event falls in the late window.

The dominance of thrust mechanisms across both windows (6/7 events) means fault mechanism type alone cannot explain the bimodal lag. However, the single mechanistic outlier (strike-slip, intra-slab) is in the late window, which may reflect either its greater depth or the fact that intra-slab stress environments couple differently to surface perturbations than interplate megathrust faults.

A more granular sub-classification is informative: among the thrust events, the early window contains only interplate megathrust events (interface slip), while the late window contains one

intra-slab thrust (Japan), one intra-slab strike-slip (Alaska), and two megathrust events (Vanuatu, Ryukyu). The Japan intra-slab thrust is itself an aftershock of the 2011 Tohoku sequence, suggesting accumulated stress within the subducting slab rather than on the primary interface.

4.3 Storm Peak Bz Intensity

Peak Bz values at storm onset differ systematically between the two lag windows:

Early-window (24–48 h): −13.9, −15.6, −13.2 nT → Mean −14.2 nT
Late-window (72–96 h): −15.5, −15.4, −15.5, −16.7 nT → Mean −15.8 nT

The late-window events occurred during storms with consistently stronger peak Bz excursions (−1.6 nT more intense on average). Every single late-window storm had a peak Bz at or below −15.4 nT; every single early-window storm had a peak Bz at or above −13.2 nT with no overlap between groups (though the ranges nearly touch at −15.6 vs −15.4).

This is the most robust pattern in the dataset. It is consistent with a graded-response model in which more intense geomagnetic storms produce longer-duration ionospheric and lithospheric disturbance, shifting the effective trigger window later. Alternatively, more intense storms may reach deeper faults (greater GIC penetration depth), naturally selecting for deeper, later-responding events. Both interpretations are speculative at this sample size but both are physically coherent.

This observation also has direct implications for prospective study design: the Bz intensity of a triggering storm should be recorded as a continuous covariate, not just a binary threshold exceedance, and lag time should be regressed against peak Bz in any future analysis with adequate sample size.

4.4 Tectonic Zone Distribution

The early-window events are geographically concentrated: two in the PNG/Solomon Islands arc and one in the Alaska Peninsula. All three are in the Pacific Ring of Fire, on the eastern and western Pacific margin respectively. The late-window events span five distinct zones (Alaska, Japan, Vanuatu, Ryukyu) and are geographically distributed around the Pacific. No obvious geographic clustering within the late window is apparent.

The Solomon Islands doublet (Hits 1 and 2, March 29 and May 7, 2015) warrants separate attention. These two events occurred 39 days apart in the same tectonic zone. Under Gardner–Knopoff (1974) parameters, the temporal aftershock window for an M7.5 event is approximately 730 days; the spatial window is approximately 81–85 km. If the May 7 M7.1 event lies within this distance of the March 29 M7.5 epicentre, it should have been declustered as an aftershock. The backfill script applies Gardner–Knopoff to the earthquake catalogue but not to the storm trigger catalogue, meaning two independent storms can both ‘hit’ the same aftershock sequence. The geographic separation of the two epicentres is not available in the backfill JSON and should be verified against the raw USGS catalogue before these are treated as fully independent hits.

5. Interpretive Hypotheses for the Bimodal Lag

The focal mechanism and depth analysis suggests three candidate hypotheses for the bimodal lag structure. These are listed in order of current evidential support, weakest to strongest, and are not mutually exclusive.

Hypothesis A: Depth-Mediated Stress Propagation Delay

Deeper events respond on longer timescales because any surface-originating stress perturbation must propagate through more lithosphere before reaching the hypocentre. Evidence for: the two deepest events (Japan 41–57 km; Alaska 60–80 km) are both in the late window. Evidence against: Vanuatu (~10–28 km) and Ryukyu (25 km) are shallow yet also in the late window. Assessment: weak directional support; not yet falsified.

Hypothesis B: Storm Intensity Governs Duration of Effective Trigger Window

More intense geomagnetic storms (−15 to −17 nT) sustain ionospheric and possibly lithospheric perturbation for longer, delaying or prolonging the stress-transfer window. Evidence for: complete rank separation between early and late windows on peak Bz (all late events ≤ −15.4 nT; all early events ≥ −13.2 nT). Evidence against: the separation is marginal and the sample size is 7. Assessment: the strongest pattern in the dataset; should be the primary covariate in prospective design.

Hypothesis C: Intra-slab vs. Megathrust Differential Response

Interplate megathrust faults respond rapidly (24–48 h) to geomagnetic perturbation because they are shallow, large, and critically stressed; intra-slab faults (deeper, less coupled to the plate interface) respond more slowly (72–96 h). Evidence for: the two confirmed intra-slab events (Japan: intra-slab thrust; Alaska: intra-slab strike-slip) are both in the late window. Evidence against: Vanuatu and Ryukyu are likely megathrust events yet are in the late window. Assessment: weak support; largely subsumed by Hypothesis A.

Hypothesis D: Random Variation (Null)

The bimodal structure is an artefact of small-sample variation. Under a uniform distribution of hits across the 72-hour window, the probability of obtaining the observed pattern (3 hits, then 0, then 4) by chance is low but not negligible for n = 7. Assessment: cannot be ruled out at current sample size. Prospective data collection will be the key test.

6. Recommendations for Prospective Study Design

Based on this analysis, the following refinements to the prospective Thesis A replication (target: 25+ new triggers at Bz ≤ −12 nT over 2–3 years) are recommended:

Record peak Bz as a continuous covariate: For each trigger event, log the exact peak Bz value (not just the threshold exceedance). Regress lag time against peak Bz across all future hits. A significant positive relationship (more negative Bz → longer lag) would directly support Hypothesis B.
Record hypocentral depth for every hit event: Depth data is available from USGS event pages and should be logged automatically. A depth vs. lag regression with n ≥ 15 hits would be powered to detect the directional signal seen here.
Classify focal mechanism for every hit event: Record rake angle from the GCMT or USGS moment-tensor solution. Classify each hit as interplate-thrust, intra-slab-thrust, or strike-slip. Stratify hit rates by mechanism type.
Add Dst < −100 nT as a parallel trigger criterion: Run the same analysis using Dst-based storm classification as an independent operationalisation of the same underlying geomagnetic disturbance. Concordance between Bz-triggered and Dst-triggered results would strengthen the observational basis.
Verify Solomon Islands doublet independence: Pull the exact epicentre coordinates for Hits 1 and 2 from the USGS ComCat and compute the inter-event distance. If < 85 km, apply the

Gardner–Knopoff rule and treat them as a single hit, reducing the exploratory n from 7 to 6 (p-value and RR should be recalculated).
Log dI/dt (rate of change) at trigger time: Geomagnetically induced currents are driven by the temporal derivative of the geomagnetic field, not the steady-state amplitude. Record the maximum dBz/dt in the 3 hours preceding the trigger threshold crossing as an additional covariate alongside peak Bz.

7. Limitations

Sample size of 7 hit events is insufficient for any statistical test of group differences. All patterns reported here are descriptive and hypothesis-generating only.
Depth estimates vary by 10–20 km between agencies for several events. The depth pattern may not survive more precise hypocenter relocation.
The USGS earthquake.usgs.gov API was not accessible from the analysis environment; focal mechanism data were retrieved from secondary sources (news reporting, institutional announcements, and review literature). Where possible, multiple sources were cross-checked.
Gardner–Knopoff declustering parameters were calibrated on California seismicity (1974) and may not accurately represent aftershock productivity or duration in global subduction zones. This limitation applies to the parent Thesis A analysis and is not specific to this addendum.
The Solomon Islands doublet (Hits 1 and 2) requires geographic verification before both events can be treated as independent observations.
No ENSO or solar cycle phase control was applied. The late-window, high-intensity storms are somewhat more likely near solar maximum; if subduction seismicity also peaks near solar maximum for other reasons, this could partially confound the Bz intensity vs. lag relationship.

8. Summary

Analysis of focal mechanisms and depths for the seven Bz ≤ −12 nT hit events reveals one strong pattern and one suggestive pattern:

> **Strong:** Storm intensity (peak Bz) is completely rank-separated between the early and late windows. All four late-window events were triggered by storms with peak Bz ≤ −15.4 nT; all three early-window events by storms with peak Bz ≥ −13.2 nT. This is the most actionable finding and motivates recording Bz as a continuous variable in prospective work.
>
> **Suggestive:** The two deepest events in the dataset (Japan 41–57 km; Alaska 60–80 km) both fall in the late window. Shallow-focus megathrust events dominate the early window. This is directionally consistent with a depth-mediated propagation delay but is not statistically significant at n = 7.

No finding in this addendum validates a causal interpretation. The bimodal lag structure may be random variation amplified by small sample size. The value of this analysis is to sharpen the prospective study design: record peak Bz, depth, and mechanism type for every future hit, and regress lag time against these covariates once n ≥ 15 hits are available.

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